# Course Syllabus

Edit

**Instructor:**

- Prof. Reza Ebadi
- rebadi@wpi.edu (mailto:aebadi@wpi.edu) (Send emails directly. Please don't send a message via Canvas)

**Assumed Background:**

- Calculus I (MA 1021),
- Calculus II (MA 1022)
- Introduction to Static Systems (ES 2501)
- Introduction to Dynamic Systems (ES 2503).

**Location and Time:**

- The lectures will be recorded and shared via Echo360

**Office Hours:**

- By appointment via Zoom. You may let me know during the lecture or via email should you need assistance with the lecture and/or assignments.

**Required Text:**

- **Design of Machinery**, 6e, Robert L. Norton, McGraw Hill Publications, ISBN 13: 9781260113310

**Course Objectives:**

Learn fundamentals of planar mechanisms, their analysis and synthesis techniques:

a. Learn to analyze mechanisms/linkages using graphical and analytical techniques.
b. Learn to synthesize mechanisms/linkages using computational techniques
c. Learn to design mechanisms

d. Improve technical communication and presentation skills

**Software:**

- We may use MATLAB (or similar software) as needed.
- You may need to use SOLIDWORKS (or any other CAD software) for linkage synthesis. These software packages are available on WPI servers.

**Grading:**

- Quizzes (20%)
- Homework assignments (20%)
- Exams (60%)

**AI Use Policy:**

**I cannot monitor how you use AI outside of assessments (I don't even want to), But I do expect you to understand the material.**

- **Allowed:**
  - Brainstorming and idea generation
  - Literature review and concept explanations
  - Code verification or optimization (must be able to explain your code)
  - Improve writing quality
- **Not Allowed:**
  - AI use during live quizzes or exams
  - AI use to find homework solutions
  - Citing AI as a source

- **This course builds foundation for what cones next:** Mechanism analysis appears in machine design, robotics, and MQP. If you skip the learning now, you'll struggle later.
- **Your future employers expect you to verify AI, not trust it blindly.** Engineers are responsible for their work. "The AI told me" is not a professional defense.
- **AI makes mistakes.** It will confidently give you wrong answers. If you don't understand the underlying concepts, you won't catch them. We'll see this together in our weekly AI dissection sessions.
- **How to use AI without sabotaging yourself:**

- Attempt problems first. Struggle is where learning happens.
    - If stuck, ask AI to explain concepts - not solve the problem for you.
    - If you use AI to check your work, make sure you can explain every step.
        - If you can't explain it, you didn't learn it.
    - **Use AI to extend beyond what we cover.** We focus on 4-bar linkages. Your project might need a 6-bar. Use your understanding of linkage analysis to learn new territory with AI's help - that's what professionals do.
- **You can outsource your thinking to AI, you cannot outsource your understanding.**

**Final Grade Letter:**

- A:≥90
- B:≥80
- C:≥70
- NR:<70

**Course Policies and Notes:**

- Homework assignments must be submitted through Canvas.
- Late homework and quizzes will not be accepted for any reason unless you have made prior arrangements with the instructor.
- Late project will be accepted with a penalty of 25 points (out of 100) per day. **No project submitted more than 2 days late will be graded.**

**Academic Honesty:**

Ethical behavior is essential for engineering professionals due to the nature of the work and is expected during your academic training as well. You are required to comply with all **University poli (https://www.wpi.edu/sites/default/files/docs/Offices/Marketing-Communications/AIStudentguide2019-2020-pages.pdf) cies (https://www.wpi.edu/sites/default/files/docs/Offices/Marketing-Communications/AIStudentguide2019-2020-pages.pdf)** regarding Academic Honesty and to familiarize yourself with those policies. Collaborating on assignments is acceptable in this course to support peer-to-peer learning. HOWEVER, you should only discuss the problems conceptually and refrain from exchanging digital copies of the assignments. Let me know if you have any questions about what constitutes academic dishonesty in this course.

**Accessibility Services for Students:**

WPI is committed to providing students with documented disabilities equal access to all university programs and facilities. If you think you have a disability requiring accommodations, you must contact **[the Office of Accessibility Services (https://www.wpi.edu/offices/office-accessibility-services)]** and get them to send me proper accommodation letters. **I will not be able to extend an exam time without accommodation letters.**

**Emotional or Mental Health Distress:**

Your academic success in this course is very important to me. If, during the term, you find emotional or mental health issues are affecting that success, please contact **WPI's Student Development (https://www.wpi.edu/offices/student-development-counseling-center) & Counseling Center (https://www.wpi.edu/offices/student-development-counseling-center)**, which provides counseling appointments and other mental health services.

**Tentative Course Outline:**

Here is a tentative class schedule. **Please note the dates/topics are subject to change.**

Week 0:

- Course Introductions

Week 1:

- Kinematics
- Degree of Freedom (DOF) and Mobility
- Number Synthesis
- Grashof Condition and Linkage Transformations

Week 2:

- Graphical Linkage Synthesis
- Two Position Rocker Output
- Two Position Coupler Output
- Three Position Coupler Output
- Quick Return Linkage
- Coupler Curves

Week 3:

- Position Analysis of Synthesized Linkages (Graphical, Analytical, Vector Loop)

Week 4:

- **Midterm Exam**
- Velocity Analysis (Graphical, Analytical, Vector Loop)
- Instant Centers of Velocity

Week 5:

- Acceleration Analysis (Graphical, Analytical, Vector Loop)
- **Final Exam**